<Final Report: EEE4610-01>

Research and Development of Electrical Bio-adhesive Materials

Report Advisor: Yejin Jo

June 2023

Seoyeon Kim, Wonjun Jang
School of Electrical and Electronic Engineering
College of Engineering
Yonsei University

Contents

Abstract ····· i
1. Introduction ····· 1
2. Hydrogel ····· 4
2.1. Polyvinyl alcohol (PVA) ····· 4
2.2. Tannic Acid (TA) ····· 5
2.3. Crosslinking effects of fabricated hydrogel-based adhesive ····· 6
3. Fabrication of the hydrogel-based adhesive ····· 7
3.1. Concentration optimization ····· 7
3.2. Thickness optimization ····· 10
3.3. Observations of the fabricated hydrogel-based adhesive's physical traits ····· 10
4. Processing and measurement for test structures ····· 13
4.1. Substrate bonding strength ····· 13
4.2. Hetero bonding strength ····· 14
4.3. Hydrogel bonding strength ····· 15
5. Applications ····· 15
5.1. Hydrogel-based stretchable Circuit ····· 15
5.2. Strain sensor fabrication ····· 17
6. Conclusion ····· 18
Reference ····· 20

Abstract

Research and Development of Electrical Bio-adhesive Materials

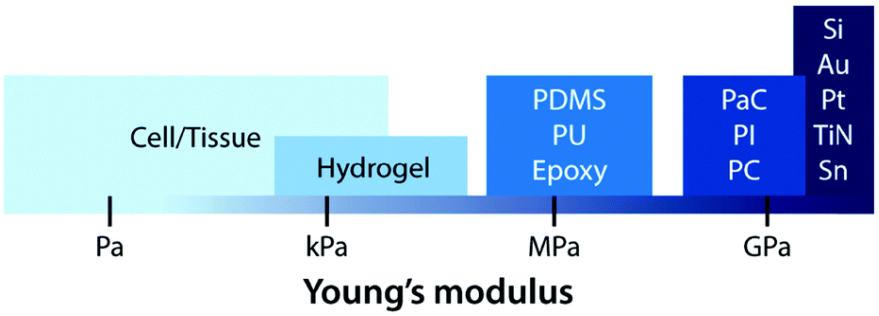
Soft electronics such as wearable devices, and implantable devices gain attention due to their characteristic for monitoring and transmitting biological signals to the tissue. These soft bioelectronics are fabricated with materials including, metallic substrates, flexible substrates, and stretchable substrates. Also, as hydrogel materials are also bringing attention as novel biocompatible materials for soft bioelectronics, a new adhesive that can overcome limitations of the conventional adhesives and previous hydrogel adhesion methods is recommended. This novel adhesive is required to provide biocompatibility, flexibility, electrical conductivity, and a conformal attachment to the surfaces of the substrates by forming reversible bonds that allow the regeneration of adhesive properties in the dynamic biological environment. In this research, we suggest a hydrogel-based adhesive fabricated with tannic acid (TA) and polyvinyl alcohol (PVA) organic compound solutions. 3 types of fabricated hydrogel-based adhesive's bonding strength measurement experiments were conducted. The highest bonding strength in nonporous substrates was 4.109 MPa for the stainless-steel sample. In porous substrates, the highest values were polyimide sample's bonding strength, 637.6 kPa. Thus, the adhesive property was well applied for the porous and nonporous substrates. For the bonding strengths measurement between hetero substrates, polydimethylsiloxane (PDMS), polyimide (PI), Ecoflex, and stainless steel (SS) were used for substrates. The highest measured bonding strength was SS-PI sample with 1296.5 kPa. The bonding strength in PI-PDM samples was 175.0 kPa. Thus, the hydrogel-based adhesive can provide sufficient adhesion properties in fabricating soft bioelectronic devices that require various types of materials. Lastly, the bonding strengths between the previous substrates (PDMS, PI, SS) with hydrogel substrates (PVA, PVA/TA, and PAAm-alginate hydrogel) were measured. For the PVA hydrogel, the highest bonding strength was 8.87kPa with PI. For PAAm-alginate, the highest bonding strength was 7.28kPa with SS.

In contrast, PVA/TA exhibited relatively higher bonding strength with the value of 50.41kPa for PI. The relatively high bonding strengths of the PVA/TA hydrogel substrate samples were due to the overlapping cross-linking interactions between the fabricated hydrogel-based adhesive and the PVA/TA hydrogel itself. These results also show the availability of the hydrogel-based adhesive that can combine the hydrogel materials with other dry, common materials used in soft bioelectronics. Then, the hydrogel-based adhesive was applied to a simple circuit and was confirmed that the adhesive maintains conductivity when stretched. The adhesive also maintained the conductivity in the strain sensor, and it was also able to successfully deliver the differentiated signals of varying joint movements.

Keywords: hydrogel-based adhesive, tannic acid, polyvinyl alcohol, soft bioelectronics, soft materials, wearable devices, implantable devices

Introduction

Soft bioelectronics gains attention due to their characteristic for monitoring and transmitting biological signals to the tissue. It can monitor electrophysiological signals (e.g. changes in voltage or current), physical signals (e.g. motion, temperature), and biochemical signals (e.g. pH, glucose, etc.) (27, 36). Especially regarding wearable and implantable soft bioelectronics, device designing, and material designing are playing a crucial role in making novel devices compatible with the human body (36). Conventional bioelectronic devices are composed of rigid, inorganic materials that mismatch with biological tissues. This may result in significant adverse effects including biological tissue damage, irritations on the skin and in tissues, and limited short lifetime of the devices (8). To overcome these issues, designing flexible, soft, stretchable devices conforming to the human body has become mainstream (40). Thus, to improve sensing ability and compatibility with the human body, three types of compatibilities should be considered: physical, biological, and interfacial compatibility.



[Figure. 1. Young's moduli of cell/tissue and common electrode materials (37)]

The biological tissues form Young's moduli in the range of 1kPa to 100kPa (27), while other rigid materials used in electronics (e.g. gold, platinum, tin, silicon, etc.) are in the GPa range (37). As Young's moduli values for the conventional electrode materials differ in several orders compared to Young's moduli of the biological tissues, efforts to develop softer materials that minimize the biomechanical dissimilarities (Young's moduli gaps) had been conducted over decades (37). Examples are polymeric materials like polyimide (PI) and elastomers like

polydimethylsiloxane (PDMS) (27,37). However, still, those Young's moduli are in the range of 1MPa to 1GPa which still have big gaps between the biological tissues (27,37) not allowing sufficient interfacial compatibility. Due to the huge differences in Young's moduli values between the biological tissues and the materials consisting of the bioelectronics, the devices lack tissue adaptability causing unwanted inflammatory responses or irritations. Additionally, for implantable electronics, a wet tissue environment may also degrade the interfacing efficiency among diverse materials used in soft bioelectronics (27). Lastly, as most bioelectronic materials maintain robust planar interfacial surfaces, this may not be conformally attached to curved, dynamically moving biological tissues (41). As a total resolution, recently hydrogel materials are recommended as physically, and biochemically compatible materials that may directly interface with the soft tissues and other previous bioelectronics materials (27, 37, 39). They can be designed to match Young's moduli of the biological tissues, and conformally attach to the interfacing surfaces due to its flexibility, and adhesive properties (27,36,37, 39). However, another issue is raised when applying hydrogel materials to soft bioelectronics. As these hydrogel materials carry abundant water, combining them with other relatively rigid, dry bioelectronic materials mentioned above is challenging. Moreover, not only combining hydrogel materials to other commonly used materials is challenging, the adhesion between the different types of materials that are mentioned above are also limited when traditional adhesives are applied.

Representative traditional adhesives that are already utilized in fabricating soft bioelectronics these days are epoxy-based adhesives, and cyanoacrylate adhesives (2-6, 35). The epoxy-based adhesives such as ACF and thiol-epoxy adhesives are used in structural bonding applications in attaching components to a printed circuit board (PCB), PI, and PDMS (2, 35). This adhesive provides strong, durable bonds and can be cured at room temperature, and has been applied to rigid and also for flexible substrates (3, 35). However, the epoxy-based adhesive has limitations of easy cracking, and facile detachment (4), such requires specific mixing and surface preparation for application, and several hours of curing which are time-consuming processes (5, 35). Another example of traditional adhesive is cyanoacrylate adhesive, which is possible for instant bonding, room-temperature-curing adhesives, and adhesive properties over a wide

variety of substrates like plastics, metals, rubber, wood, and skin (6). However, it still has the limitations of low flexibility, brittleness, and limited biocompatibility regarding living tissues other than external skin tissues. Moreover, it has low electrical conductivity and thus lacks the availability of electrical connections or electrophysiological signals transmission (6). Due to these limitations, these traditional adhesive materials not only are unable to combine the hydrogel materials with other materials but also are unable to provide promising adhesive properties when making soft bioelectronics.

Thus, regarding the adhesion of the hydrogel materials, several studies have worked on methods that allow the hydrogel materials to adhere to substrates that are porous, nonporous, hydrophilic, and hydrophobic materials. For example, Yuk, H. *et al.* used the anchoring method making use of covalent bonding crosslinking of hydrogel tough bonding networks onto the nonporous solid surfaces including glass, silicon, ceramics, titanium, and aluminum (8). However, this adhesive method requires surface functionalization before adhesive bonding is done (8) which remains a limitation of choosing optimal anchoring chemical compounds for specific substrates. Another alternative method of using ultraviolet-initiated polymerization (11) was also recommended by Yuk, H. *et al.*, but this also requires extra curing processes that have challenges to sensitive target surfaces (9).

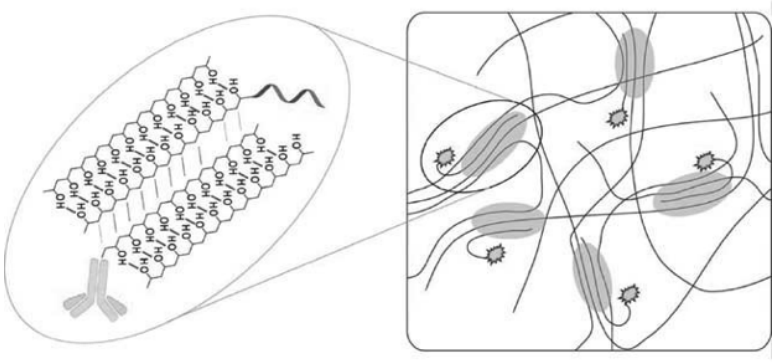
Since several limitations are remaining in conventional adhesives and recent studies on adhering hydrogel materials to other common soft bioelectronics materials, a novel adhesive is required for future applications in soft bioelectronics, and this adhesive should be satisfying such conditions. The adhesive should be providing biocompatibility, flexibility, electrical conductivity, and a conformal attachment to the surfaces of the substrates by forming reversible bonds that allow the regeneration of adhesive properties in the dynamic biological environment. This research thus recommends the hydrogel-based adhesive as an ultimate resolution regarding hydrogel materials' adhesive properties toward diverse kinds of substrates including porous substrates (PI, polypropylene (PP), acrylic (Acryl), PDMS), and nonporous substrates (gold, platinum, tin, silicon, etc.), and adhesive properties between the common materials mentioned above. Our research targets the fabrication of a versatile hydrogel-based adhesive that can generate adhesion incidentally with diverse kinds of substrates included in soft electronics,

especially for the rigid chip, flexible substrates, stretchable substrates, and hydrogel substrates. Also, we aim to fabricate a hydrogel-based adhesive that can conformal and interfacial compatibility over any kind of substrate surfaces including nonporous materials. Moreover, we are planning to take advantage of the hydrogel material that has biocompatibility which is necessary for soft bioelectronics. To cover these features at once, we have chosen polyvinyl alcohol (PVA) and tannic acid (TA) as the base materials for hydrogel glue which has the physical, and chemical properties that can contribute to mentioned objectives of the hydrogel-based adhesive.

2. Hydrogel characteristics

As we are planning to make a hydrogel-based adhesive, we focused on the two factors of cohesion and adhesion that determine the adhesiveness of hydrogel-based adhesive.

2.1. Polyvinyl alcohol (PVA)

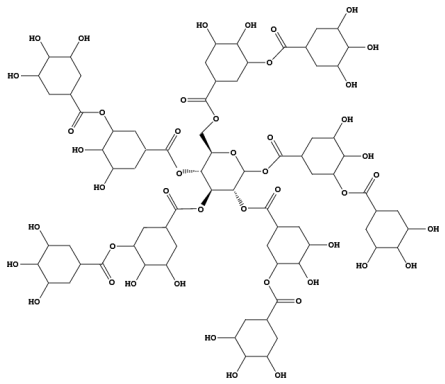


[Figure. 2. Association of PVA polymer chains in materials design (13)]

PVA (Polyvinyl alcohol) is a synthetic polyhydroxy polymer that dissolves well in water due to its polarity thanks to hydroxyl functional groups attached to its main chain (14,15). Its abundant hydroxyl groups are targeted in chemical modifications in diverse applications, and especially, cross-linking of PVA is attained in soft bioelectronic material development (16). The

cross-linking is usually attained both in chemical and physical methods when the PVA is used in material design (17). This polymer has preferable mechanical features in strength, and flexibility, and is typically suitable for bioelectronics thanks to its biocompatibility (18). Additionally, when PVA is in a state of good saponification and stereoregularity, it exhibits high strength and elasticity, making it suitable for producing hydrogels at appropriate concentrations. As the PVA concentration increases, Young's moduli decrease monotonically. On the other hand, the adhesive force increases with good uniformity and stretchability (19).

2.2. Tannic Acid (TA)

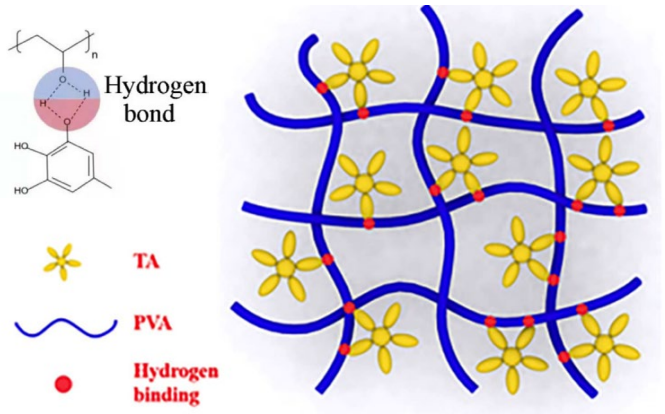


[**Figure. 3. Tannic acid molecule structure (20)**]

Tannic Acid (TA) is one of the polyphenols that is widely used in various industrial fields, including medicine because it is harmless and can bond with many oxidants. TA contains a sufficient number of -OH groups. It contains five pyrogallol and five catechol groups, thus total 25 of hydroxyl functional groups (33), which makes it highly soluble in water but also prone to self-aggregation (21). Also, these functional groups provide multiple bonding sites through intermolecular interactions such as hydrogen bonding, hydrophobic interactions, and ionic bonds (33). For example, phenolic functional groups (catechol/pyrogallol functional groups) can form coordinates with metal ions (34). In addition, hydroxyl functional groups participating in hydrogen bonding play a major role in adhesion (33,34). For example, the active groups of

TA can react with nucleophiles such as amide bonds, amines, and thiols of the substrate surfaces (34). Therefore, TA may contribute to adhesive performances based on these intermolecular interactions and TA may also facilitate cross-linking with PVA (22) which we will further explain in the paper.

2.3. Crosslinking effects of fabricated hydrogel glue



[**Figure. 4 Cross-linking of PVA main chain with TA through hydrogen bonding (23, 24)**]

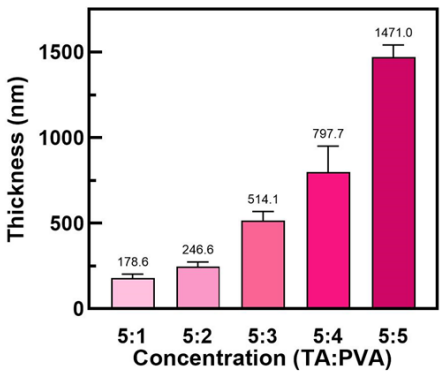
The hydrogel glue fabricated in this research uses the self-assembly between PVA and TA. As mentioned above TA is possible for physical crosslinking as well as chemical crosslinking due to its number of functional groups. Hydrogen bonding is one of the most basic physical interactions occurring between polyphenol molecules allowing the adhesion ability (25). Thus, PVA and TA are interacting through hydrogen bonding, and a large number of phenyl groups in TA allows multiple coupling to a vast amount of PVA molecules' hydroxyl groups. This in return generates hydrogel with multidentate hydrogen bonding formed through this catechol or pyrogallol group which is stronger than a single-ligand hydrogen bonding (25). Thus, PVA-TA hydrogel shows two kinds of cross-linking. First, chemical cross-linking in the long chain of PVA is the main chain. Secondly, the hydrogen bonding between the polymer (PVA) chain and TA molecules (26). Especially for the second cross-linking contributes to energy dissipation leading to improved adhesion (8, 27). Considering these features of the PVA and TA, our

research expects to fabricate hydrogel glue enabled to toughly bond to materials used in soft electronics including rigid or flexible, stretchable substrates, or for hydrogel substrates.

3. Fabrication of the hydrogel-based adhesive

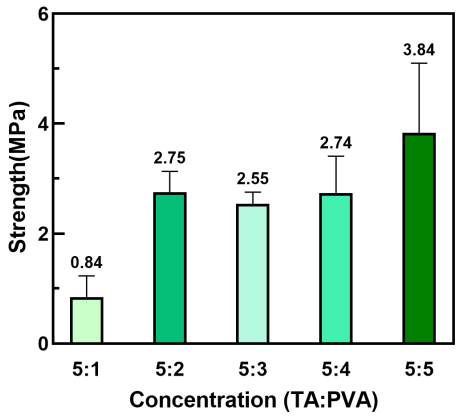
3.1 Concentration optimization

Before optimizing the total concentration of the TA and PVA solutions, we first conducted experiments deciding the ratio of the concentration of the TA solution and PVA solution. As mentioned above, 510mg of TA or PVA in 20ml DI water is set as the standard concentration of 100%, and we gradually decreased the concentration of the PVA solution. The layer coating was conducted in the order of TA-PVA-TA-PVA-TA. The reason to put the TA layer as the first layer was because a TA compound has a number of hydroxyl groups (25 hydroxyl groups). This fact allows the TA adhesive layer to more easily attach to the substrate that is being coated compared to the PVA adhesive layer. In the same context, the layer of the hydrogel-based adhesive is finished with the TA adhesive layer. This concept is used throughout the experiments conducted in this research.



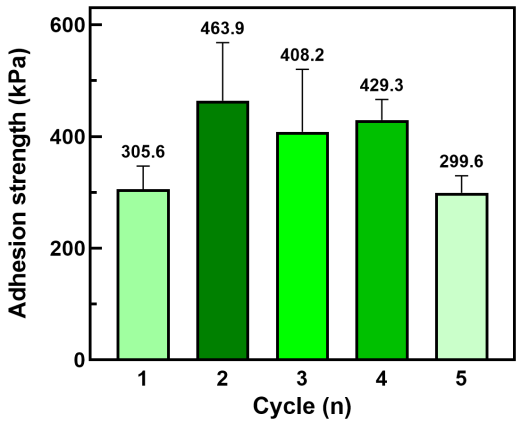
[Figure. 5. Thickness of the hydrogel-based adhesive layers versus concentration of TA: PVA]

As shown in Fig.7, we can see that the thickness of the hydrogel-based adhesive increased gradually as the concentration of the PVA solution increased from 20% to 100%. Also, the thickness of the 100% concentration hydrogel-based adhesive exceeded the target thickness, 1μm, implying that the optimal concentration would be below 100%. However, as we should also be considering the second variable, the bonding strength, we decided on the optimal ratio of the concentration of TA solution over PVA solution after examining the tensile tests.



[Figure. 6. Bonding strength of the hydrogel-based adhesive layers versus concentration of TA: PVA]

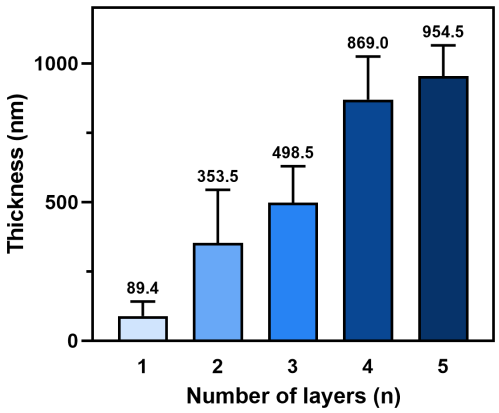
Our research conducted tensile tests to examine the bonding strengths of the hydrogel-based adhesive with varying concentrations of PVA solutions. From Fig.7, the total thickness of the hydrogel-based adhesive of the 5:5 ratio concentration exceeded 1μm while the thickness of the 5:4 ratio was around 0.8μm. However, as we obtained the bonding strength data, we realized that the strength of the 5:5 ratio concentration hydrogel-based adhesive exhibited a noticeable strength of 3.84 MPa among other ratio concentrations. Therefore, we concluded that for better adhesive properties, the concentration ratio of the TA and PVA solutions should be equal.



[Figure. 7. Adhesion strength over coating cycle (1 cycle = TA 1+ PVA 1)]

After finding the optimal concentration ratio of the TA and PVA solution as 5:5, we next conducted an experiment to decide the number of repetitions of the coating. We conducted adhesion tests regarding the number of cycles. One cycle in the figure refers to 1 layer of TA adhesive and 1 layer of PVA adhesive. From Fig.9, the adhesion strength value was the highest with 463.9 kPa when the cycle was done twice. Therefore, we chose the optimal coating cycle as a 2-cycle. In addition, as we know the TA chemical structure has 25 hydroxyl groups, we expected that by additionally adding the TA layer as the last layer the adhesive property would be enhanced thanks to the active hydrogen bonding interactions of the TA. Thus, we concluded that the optimal adhesive layer is TA-PVA-TA-PVA-TA.

3.2 Thickness optimization

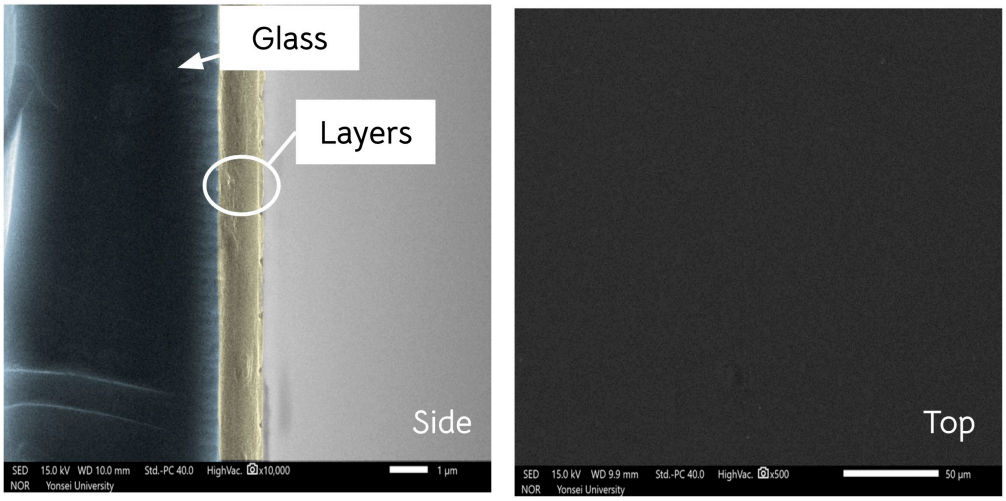


[**Figure. 8. Thickness in the number of layers of hydrogel-based adhesives**]

This experiment was conducted to confirm that the thickness of the hydrogel-based adhesive was below 1um. We alternated layers of TA adhesive and PVA adhesive to measure the change in thickness depending on the number of layers. The results showed that when 5 layers of TA adhesive and PVA adhesive were coated the thickness was close to 1 um but still below. Hence, we confirmed that fabricating hydrogel-based adhesive with five layers is possible.

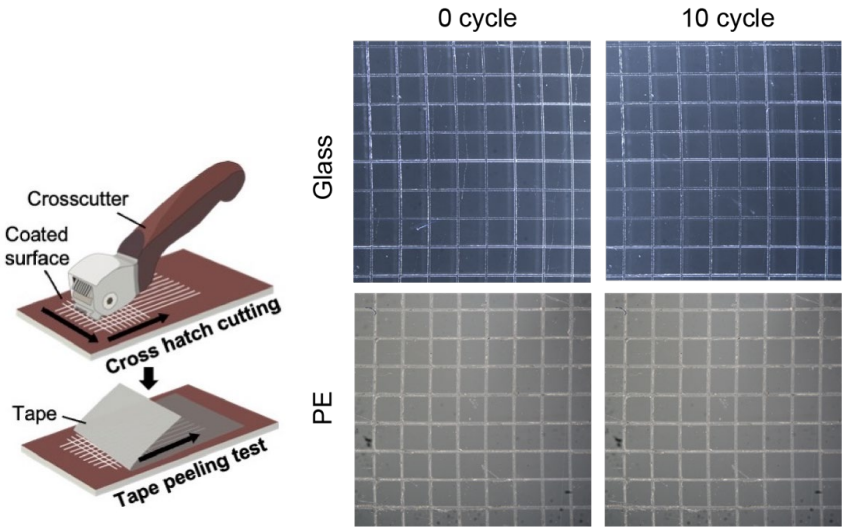
3.3 Observations of the fabricated hydrogel-based adhesive's physical traits

To observe how the hydrogel-based adhesive layers interact with each other we observed the scanning electron microscope (SEM) images of the side view of the substrate coated with the hydrogel-based adhesive. Also, to confirm that the hydrogel-based adhesive is uniformly distributed, we observed the top view of the coated substrate. Then, we examined the cross-cut test 0 to 10 cycles to confirm that the adhesive layers are connected well to each other and also attached to the substrate's surface.



[**Figure. 9. SEM images of the side & top view of the hydrogel-based adhesive layers**]

The side views of the SEM image show how the hydrogel-based adhesive layers are formed. The 5 layers weren't able to discern since the boundaries were blurred. We analyzed that this was because of the cross-linking effect between the TA and PVA. Since the two compounds actively interact with each other through hydrogen bonding, the boundaries of the layers seemed to be fused together. Through this observation, we concluded that cross-linking effect contributed to the hydrogel-based adhesive's stronger bonding property. From the top view SEM image, we confirmed that the adhesive itself had distributed uniformly after the coating process.

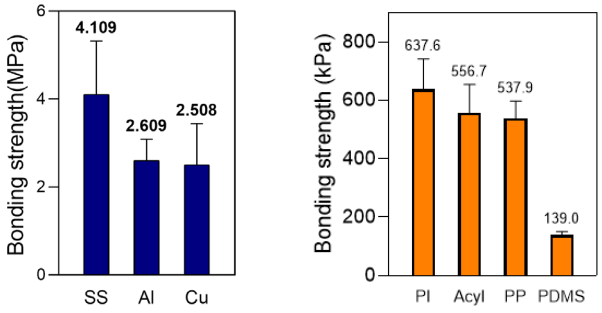


[**Figure. 10. Representative optical images of coating layers after the cross-cut test (42)**]

The 0 cycle in the cross-cut image refers to the one performance of a cross-cutting of the substrates' coated surface with the cross-cutter. Then, the tape peeling test was repeated 10 times which is shown in the 10-cycle figures. As we can see in the Fig10-b, the coating layers had been attached robustly to the substrates as there's no peeling occurred. We concluded that the fabricated hydrogel-based adhesive can be robustly coated to the substrates.

4. Hydrogel-based adhesive characterization

4.1 Substrate bonding strength



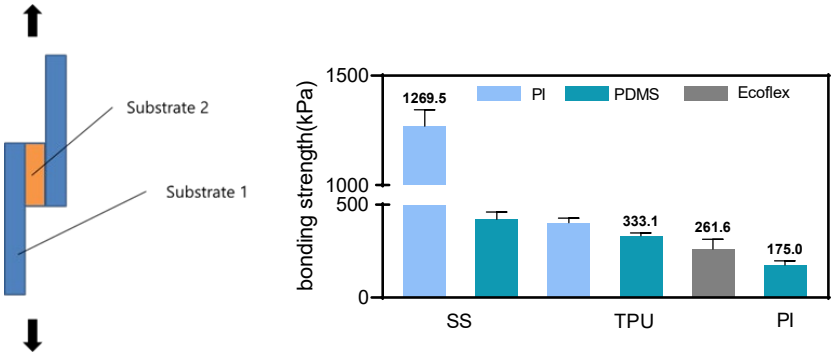
[**Figure. 11. Bonding strengths of the nonporous and porous substrates**]

The following results are from measurements of bonding strength depending on the substrate. Each substrate was coated with TA-PVA-TP-PVA-TA. In the case of nonporous substrate, apply heat for 1minute at 130°C while overlapping the substrate. In another case, apply heat for 30 minutes at 100°C while overlapping the substrate. The bonding strength was measured using a shear test. The graph on the left shows the results of experiments on nonporous substrates. The stainless steel (SS) has a bonding strength 4.109kPa which is a higher bonding strength compared to Al and Cu samples. This suggests that hydrogel-based adhesive has a stronger adhesion with SS. Since SS has a passive layer, the surface always remains uniform which allows the hydrogel-based adhesive to conformally distribute and robustly attach to the coated substrates.

The graph on the right shows the results of experiments on porous substrates. PI demonstrated a higher bonding strength of 637.6 kPa among them. PDMS had a lower bonding strength of 139.0Kpa. PDMS was strongly bonded to a hydrogel-based adhesive, but the sample was broken during the measurement process. Therefore, since the bonding strength value at the time of fracture was measured, the value was relatively low. Porous materials have numerous tiny holes, or "pores," that can increase the overall surface area in contact with the adhesive. As a result, the adhesive can penetrate into these pores, creating a larger area for bonding, which may result

in stronger adhesion (38).

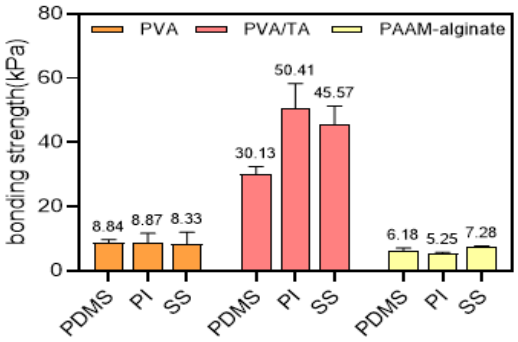
4.2 Hetero bonding strength



[**Figure. 12. Bonding strengths between hetero substrates**]

We attempted to adhere materials (2cm x 1.5cm x 0.1cm) with different properties using a hydrogel-based adhesive while applying a heat at 100°C for 30 min and measured the bonding strengths. The shear tests were performed by the tensile test machine at a rate of 50mm/min and the maximum load was recorded and then the mean values were obtained. We used PDMS, TPU, PI, Ecoflex, and SS. As a result of the experiment, the highest bonding strength was the SS-PI sample which was measured as 1269.6 kPa. Same as in section 4.1 results, the bonding strengths between SS and PI were the highest.

4.3 Hydrogel bonding strength



[**Figure. 13. Bonding strengths between the hydrogel substrate and other substrates**]

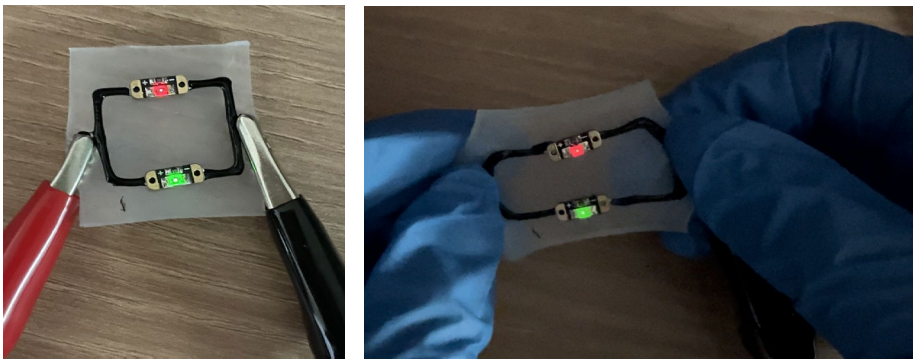
Substrates (PDMS, PI, SS) have adhered to hydrogels (PVA, PVA/TA, PAAm-alginate) using a hydrogel-based adhesive and then subjected to a test by inserting a hydrogel chip between two hydrogels, like hetero bonding. It was made in 100°C for 30 min. The experimental results show that when the hydrogel is PVA/TA, it has a significantly higher bonding strength to the factor of 3.4 to 9.6, compared to the other two experimental groups. This is due to the benzene rings of TA forming a strong bond with hydrophobic surfaces and the hydroxyl groups of TA forming a strong bond with hydrophilic surfaces, both of which greatly influence the bonding strength. Therefore, when the hydrogel is TA/PVA, unlike other hydrogels, it has a high bonding strength for various substrates.

5. Applications

5.1 Hydrogel-based stretchable Circuit

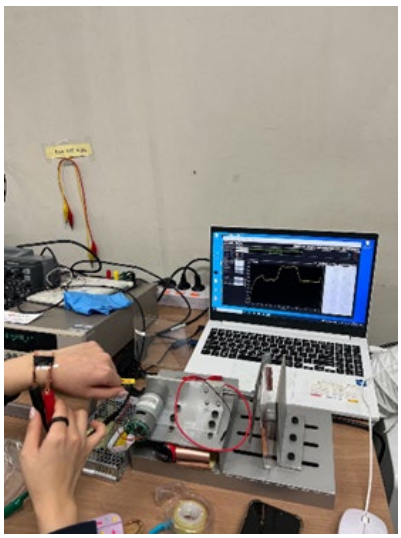
Using a 3D printer, we designed a simple circuit as shown above. We connected CNT-based wires with LEDs coated with TA-PVA-TA layers as shown in the figure 14. We applied a 3V voltage to check for conductivity in the hydrogel-based adhesive. As seen in the figure 14. we

confirmed that the LEDs were on. Then, when the substrate was stretched, the LEDs were still on. From this, we concluded that even though TA and PVA themselves do not have electrical conductivities, they may exhibit conductivities thanks to the hydrogel-based adhesive's thinness and porousness. This allows us to confirm the potential for using a hydrogel-based adhesive in soft bioelectronics. However, in this experiment, as we stretched the substrate in both directions, the light intensity of the LEDs decreased. This was due to the insufficient performance of the CNT-hydrogel. Thus, in further experiments, using functionalized CNT when fabricating the hydrogel used for the circuit wires would allow better conductivity to the simple circuit. After these conditions are satisfied, we will be able to confirm the adhesive's conductivity more accurately.



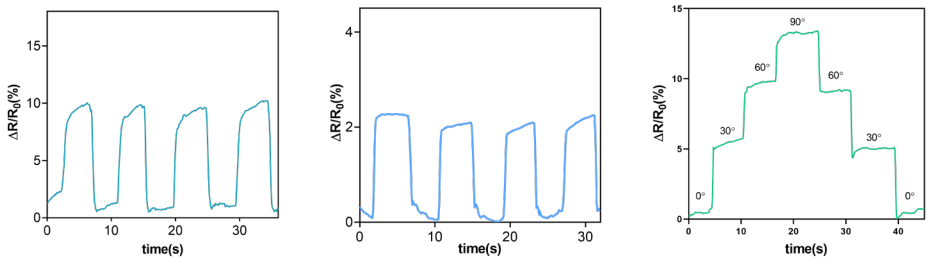
[**Figure. 14. Conductive behavior of a hydrogel-based adhesive in LED connections**]

5.2 Strain sensor fabrication



[**Figure. 15. Design and implementation of hydrogel-based strain sensor**]

We created a strain sensor with a Si-based stretchable substrate coated with TA-PVA-TA-PVA-TA, the optimized layer condition. This substrate is used as the first layer substrate that directly contacted the skin. Then the CNT-gel substrate layer was attached to the first layer which took the role of the strain sensor through the making signals of resistance change. Since the CNT-gel carries moisture, there was no additional process of swelling. After attaching the CNT-gel substrate to the Si-based stretchable substrate, we attached copper tape to both ends as shown in the picture, and applied voltage. This sensor was attached to joints such as fingers and wrists and was designed to measure the change in resistance caused by applying stress.



[**Figure. 16. Changes in resistance across varying degrees of bend in a sensor**]

Fig16-a and Fig16-b alternately present results in the initial state and the state after bending the finger and wrist, where the sensor is attached, to 90 degrees. The difference between the two states can be clearly observed through the resistance change, and it is confirmed that they restore well to the initial state even after the repetition of the measurements. Fig16-c shows the results of measurements in four different states at 0 degrees, 30 degrees, 60 degrees, and 90 degrees, after attaching the sensor to the hand. As the angle increases, the change in resistance becomes larger, and this can be clearly observed, and it is also confirmed that it restores well in the same context of Fig16-a and Fig16-b. From this result, we concluded that the strain sensor not only can sense the joint movement but also can differentiate the signals for varying joint movements. From these results, we have learned that hydrogel-based adhesive can enable connections between flexible, stretchable substrates, and maintain the electrical conductivity of the devices. Thus, our hydrogel-based adhesive can be applied to wearable soft bioelectronics.

6. Conclusion

Our research developed a hydrogel-based adhesive as an adhesive targeting wearable and implantable soft bioelectronics. It was fabricated by repeatedly coating TA adhesive layers and PVA adhesive layers. With this adhesive, we can provide biocompatibility, flexibility, allowing the current to flow, and conformal attachment to the surfaces of the substrates with the cross-linking effects that allow the regeneration of adhesive properties in the dynamic biological environment. With 3 types of bonding strength measuring experiments, we demonstrated high

bonding strengths when the hydrogel-based adhesive was applied to both nonporous and porous substrates. Additionally, the hydrogel-based adhesive showed high bonding strengths when applied in between the various substrates, making it suitable for usage in soft bioelectronic devices that requires diverse material contacts. Lastly, when applied to a PVA/TA hydrogel, the adhesive exhibited exceptionally strong bonding strength. The relatively high bonding strengths of the PVA/TA hydrogel substrate samples were due to the overlapping cross-linking interactions between the fabricated hydrogel-based adhesive and the PVA/TA hydrogel itself.

To demonstrate the applicability of hydrogel-based adhesive in wearable soft bioelectronics, we created a simple circuit using CNT-gel and two LEDs to test the conductivity of the hydrogel-based adhesive. When we applied voltages to the simple circuit, the LEDs lit up, and even when we stretched the simple circuit, the LEDs continued to emit light. This confirmed that the adhesive allowed to flow current for its thinness and porousness, and it demonstrated the potential for application in wearable soft bioelectronics. We then created a strain sensor by attaching CNT-gel to a Si-based stretchable substrate using a hydrogel-based adhesive. After attaching the strain sensor to a joint, we observed changes in the sensor's characteristics. The changes were represented accurately on a graph as we manipulated the joint's angle. This confirmed that the hydrogel-based adhesive not only provides adhesion properties but also allows the current flow, enabling precise differentiated sensing of joint movements. Therefore, the hydrogel-based adhesive is suitable for application in wearable soft bioelectronics.

Reference

[1] Park, J., Lee, Y., Kim, T. Y., Hwang, S., & Seo, J. (2022). Multifunctional Metal-Organic Frameworks for Advanced Electronics. ACS Applied Electronic Materials, 4(4), 1449-1468. DOI: 10.1021/acsaelm.1c01212

[2] Ren, H.-M., Guo, Y., Huang, S.-Y., Zhang, K., Yuen, M. M. F., Fu, X. Z., Yu, S., Sun, R., & Wong, C.-P. (2015). One-Step Preparation of Silver Hexagonal Microsheets as Electrically Conductive Adhesive Fillers for Printed Electronics. ACS Applied Materials & Interfaces, 7(24), 13685-13692.

[3] Li, C., Li, Q., Long, X., Li, T., Zhao, J., Zhang, K., E, S., Zhang, J., Li, Z., & Yao, Y. (2017). In Situ Generation of Photosensitive Silver Halide for Improving the Conductivity of Electrically Conductive Adhesives. ACS Applied Materials & Interfaces, 9(34), 29047-29054.

[4] Ma, R., Kwon, S., Zheng, Q., Kwon, H. Y., Kim, J. I., Choi, H. R., & Baik, S. (2012). Carbon-Nanotube/Silver Networks in Nitrile Butadiene Rubber for Highly Conductive Flexible Adhesives. Advanced Materials, 24(24), 3344-3349.

[5] W. Broughton. (2012). 7 - Assessing the moisture resistance of adhesives for marine environments. Adhesives in Marine Engineering. Woodhead Publishing. 155-186.

[6] Coover, H.W., Dreifus, D.W., O'Connor, J.T. (1990). Cyanoacrylate Adhesives. In: Skeist, I. (eds) Handbook of Adhesives. Springer, Boston, MA.

[7] Zhao, R., Bobev, S., Krishna, L., Yang, T., Weller, J. M., Jing, H., & Chan, C. K. (2017). ACS Applied Materials & Interfaces, 9(47), 41246-41257.

[8] Yuk, H., Zhang, T., Lin, S., et al. (2016). Tough bonding of hydrogels to diverse nonporous surfaces. Nature Materials, 15, 190-196.

[9] Wirthl, D., Pichler, R., Drack, M., Kettlguber, G., Moser, R., Gerstmayr, R., Hartmann, F., Bradt, E., Kaltseis, R., Siket, C. M., Schausberger, S. E., Hild, S., Bauer, S., & Kaltenbrunner, M. (2017). Instant tough bonding of hydrogels for soft machines and electronics. Science advances, 3(6), e1700053.

[10] Annabi, N., Tamayol, A., Uquillas, J. A., Akbari, M., Bertassoni, L. E., Cha, C., ... & Weiss, A. S. (2014). 25th anniversary article: rational design and applications of hydrogels in

regenerative medicine. Advanced Materials, 26(1), 85-123

[11] Olabi, A. G., Abdelkareem, M. A., Wilberforce, T., & Sayed, E. T. (2021). Application of graphene in energy storage device – A review. *Renewable and Sustainable Energy Reviews*, 135, 110026. https://doi.org/10.1016/j.rser.2020.110026

[12] Jiang, S., Liu, S., & Feng, W. (2011). PVA hydrogel properties for biomedical application. Journal of the Mechanical Behavior of Biomedical Materials, 4(7), 1228-1233. ISSN 1751-6161.

[13] Hassan, C. M., Peppas, N. A., & Biopolymers, P. V. A. (2000). hydrogels, anionic polymerisation, Nanocomposites. Springer, 153, 37.

[14] Wang, M., Bai, J., Shao, K., Tang, W., Zhao, X., Lin, D., Huang, S., Chen, C., Ding, Z., & Ye, J. (2021). Poly(vinyl alcohol) hydrogels: The old and new functional materials. International Journal of Polymer Science, 2021, 2225426.

[15] Sun, X., Agate, S., Salem, K. S., Lucia, L., & Pal, L. (2021). Hydrogel-based sensor networks: Compositions, properties, and applications - A review. ACS Applied Bio Materials, 4(1), 140-162.

[16] Hassan, C. M., & Peppas, N. A. (2000). Structure and applications of poly(vinyl alcohol) hydrogels produced by conventional crosslinking or by freezing/thawing methods. In Biopolymers - PVA Hydrogels, Anionic Polymerisation Nanocomposites; Advances in Polymer Science, Vol. 153 (pp. 37-65). Springer.

[17] Hassan, C. M., Peppas, N. A., & Biopolymers, P. V. A. (2000). hydrogels, anionic polymerisation, Nanocomposites. Springer, 153, 37.

[18] Sun, X., Agate, S., Salem, K. S., Lucia, L., & Pal, L. (2021). Hydrogel-Based Sensor Networks: Compositions, Properties, and Applications - A Review. ACS Applied Bio Materials, 4(1), 140-162.

[19] Liu, T., Jiao, C., Peng, X., Chen, Y.-N., Chen, Y., He, C. et al. (2018). Super-strong and tough poly(vinyl alcohol)/poly(acrylic acid) hydrogels reinforced by hydrogen bonding. *Journal of Materials Chemistry B*, 6(48), 8105–8114.

[20] Lu, Ruofei & Zhang, Xiaoqiang & Cheng, Xinxiu & Zhang, Yagang & Zan, Xingjie & Zhang, Letao. (2020). Medical Applications Based on Supramolecular Self-Assembled Materials From Tannic Acid. Frontiers in Chemistry. 8. 583484.

[21] Annabi, N., Tamayol, A., Uquillas, J. A., Akbari, M., Bertassoni, L. E., Cha, C., ... & Weiss, A. S. (2014). 25th anniversary article: rational design and applications of hydrogels in regenerative medicine. Advanced Materials, 26(1), 85-123

[22] *Chem. Educ.* (1934), 11, 12, 670 Publication.

[23] Yang, N., Yuan, R., You, D., Zhang, Q., Wang, J., Xuan, H., & Ge, L. (2022). Gallol-based constant underwater coating adhesives for severe aqueous conditions. Colloids and Surfaces A: Physicochemical and Engineering Aspects, 634, 127948.

[24] Si, C., Tian, X., Wang, Y., Wang, Z., Wang, X., Lv, D., Wang, A., Wang, F., Geng, L., Zhao, J., Hu, R., & Zhu, Q. (2022). A Polyvinyl Alcohol-Tannic Acid Gel with Exceptional Mechanical Properties and Ultraviolet Resistance. Gels (Basel, Switzerland), 8(11), 751.

[25] Chen, C., Yang, H., Yang, X., & Ma, Q. (2022). Tannic acid: a crosslinker leading to versatile functional polymeric networks: a review. RSC Advances, 12, 7689-7711.

[26] Fan, H. (2018). Tough, swelling-resistant, self-healing, and adhesive dual-cross-linked hydrogels based on polymer-tannic acid multiple hydrogen bonds. Macromolecules, 51(5), 1628-1636.

[27] Park, J., Kim, J. Y., Heo, J. H., Kim, Y., Kim, S. A., Park, K., Lee, Y., Jin, Y., Shin, S. R., Kim, D. W., & Seo, J. (2023). Intrinsically Nonswellable Multifunctional Hydrogel with Dynamic Nanoconfinement Networks for Robust Tissue-Adaptable Bioelectronics. Advanced science (Weinheim, Baden-Wurttemberg, Germany), 10(12), e2207237.

[29] Mohammed, Azad. (2018). Scanning electron microscopy (SEM): A review.

[30] Scanning electron microscopy, physics of image formation and microanalysis, L. Reimer, Springer

[31] Azom. (n.d.). High-Power DUV-LED Light Engine – 280nm to 320nm. AZoM.

[32] Abadias, G., Chason, E., Keckes, J., Barna, P. B., & Movchan, B. A. (2018). Stress in thin films and coatings: Current status, challenges, and prospects. Journal of Vacuum Science & Technology A, 36(1), 010801.

[33] Gwak, M. A., Hong, B. M., Seok, J. M., Park, S. A., & Park, W. H. (2021). Effect of tannic acid on the mechanical and adhesive properties of catechol-modified hyaluronic acid hydrogels. International Journal of Biological Macromolecules, 191, 699-705.

[34] Chen, C., Yang, H., Yang, X., & Ma, Q. (2022). Tannic acid: a crosslinker leading to versatile functional polymeric networks: a review. RSC Advances, 12, 7689-7711.

[35] Hoang, M. V., Chung, H. J., & Elias, A. L. (2016). Irreversible bonding of polyimide and polydimethylsiloxane (PDMS) based on a thiol-epoxy click reaction. Journal of Micromechanics and Microengineering, 26(10), 105019.

[36] Sunwoo, S.-H., Ha, K.-H., Lee, S., Lu, N., & Kim, D.-H. (2021). Wearable and Implantable Soft Bioelectronics: Device Designs and Material Strategies. Annual Review of Chemical and Biomolecular Engineering, 12, 359-391.

[37] Yuk, H., Lu, B., & Zhao, X. (2019). Hydrogel bioelectronics. Chemical Society Reviews, 48(6).

[38] Woodhead Publishing. (2009). Cellular Response to Biomaterials. In Woodhead Publishing Series in Biomaterials (pp. 560-571)

[39] Chaudhuri, O., Gu, L., Klumpers, D., Darnell, M., Bencherif, S. A., Weaver, J. C., Huebsch, N., Lee, H. P., Lippens, E., Duda, G. N., & Mooney, D. J. (2016). Hydrogels with tunable stress relaxation regulate stem cell fate and activity. Nature materials, 15(3), 326–334.

[40] Park, J., Lee, Y., Kim, T. Y., Hwang, S., & Seo, J. (2022). Functional Bioelectronic Materials for Long-Term Biocompatibility and Functionality. ACS Applied Electronic Materials, 4(4), 1449-1468.

[41] Lacour, S. P., Courtine, G., & Guck, J. (2016). Materials and technologies for soft implantable neuroprostheses. Nature Reviews Materials, 1(10), 16063.

[42] Park, K., Kim, S., Jo, Y., Park, J., Kim, I., Hwang, S., Lee, Y., Kim, S. Y., & Seo, J. (2023). Lubricant skin on diverse biomaterials with complex shapes via polydopamine-mediated surface functionalization for biomedical applications. Bioactive Materials, 25, 555-568.